$I_b = \frac{(-7i)I_a}{3-7i}$ $(11.55\angle 49.5)$

$10 - 21i + (7-7i)$

$5-3i =$

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$10\times$

$(1.65\angle -45)\dot{I}_a$ $15.2\angle 4.4$

16.1

$\dot{I}_1 + \dot{I}_2 - \dot{I}_3 - \dot{I}_4 = 0$

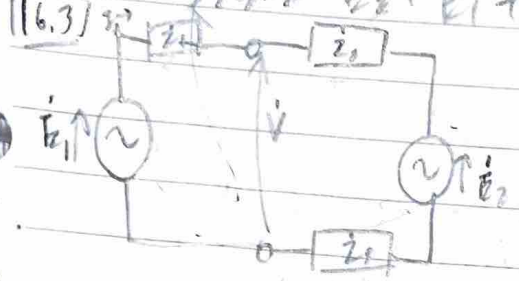
$\dot{I}_4 = \dot{I}_1 + \dot{I}_2 - \dot{I}_3 = 20\angle 0° + 10\angle 30° - 8\angle -20° = 22.5\angle 20.1°$

16.2

$-\dot{V} - \dot{Z}_2\dot{I}_2 - \dot{E}_2 + \dot{E}_1 + \dot{I}_1\dot{Z}_1 = 0$

$\dot{V} = -\dot{Z}_2\dot{I}_2 - \dot{E}_2 + \dot{E}_1 + \dot{I}_1\dot{Z}_1 =$

16.3

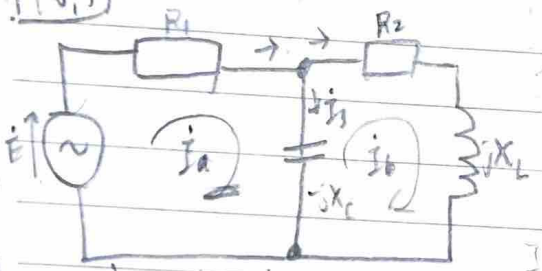


$\dot{E}_1 - \dot{E}_2 = \dot{Z}_1\dot{I} + \dot{Z}_2\dot{I} + \dot{Z}_3\dot{I}$

$\dot{I} = \dot{E}_1 - \dot{E}_2 / \dot{Z}_1 + \dot{Z}_2 + \dot{Z}_3 = 0.322\angle -37.3°$

$\dot{V} = \dot{E}_1 - \dot{Z}_1\dot{I} = 7.69\angle 14.7°$

16.4



$\dot{I}_1 = \dot{I}_2 + \dot{I}_3$, $\dot{I}_3 = \dot{I}_1 - \dot{I}_2$ ⓐ

$\dot{E} - (R_1 - jX_C)\dot{I}_a + (-jX_C)\dot{I}_b = 0$

$\dot{E} = (7-7i)I_a + (7i)I_b$ ①

$(R_2 + jX_L)\dot{I}_b - (-jX_C)\dot{I}_a = 0$

$(3+j4-j7)\dot{I}_b + (7i)I_a = 0$ ②

①、②式、$I_a = 0.66\angle -4.4°$

$I_b = 1.1\angle -49.4°$

$I_1 = I_a = 0.66\angle -4.4°$

$I_2 = I_b = 1.1\angle -49.4°$

ⓐ式、$I_3 = I_1 - I_2 = I_a - I_b = 0.79\angle 94.3°$

$\dot{Z} = \dot{E}/I_1 = 15.2\angle 4.4°$

$P_2 = I_2 \times I_2 R_2 =$

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$\boxed{16.5}$

$I_1+I_2-I_3=0$

$E_1-(j\omega L)I_1-\left(\frac{1}{j\omega C}\right)I_3=0$

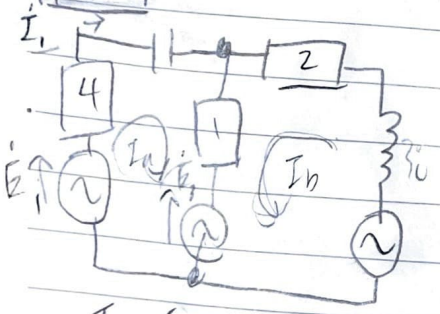
$E_2-RI_2-\left(\frac{1}{j\omega C}\right)I_3=0$

$$\begin{pmatrix} 1 & 1 & -1 \\ j\omega L & 0 & \frac{1}{j\omega C} \\ 0 & R & \frac{1}{j\omega C} \end{pmatrix}\begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix}=\begin{pmatrix} 0 \\ E_1 \\ E_2 \end{pmatrix}$$

$$I_1=\frac{\begin{vmatrix} 0 & 1 & -1 \\ E_1 & 0 & \frac{1}{j\omega C} \\ E_2 & R & \frac{1}{j\omega C} \end{vmatrix}}{\begin{vmatrix} 1 & 1 & -1 \\ j\omega L & 0 & \frac{1}{j\omega C} \\ 0 & R & \frac{1}{j\omega C} \end{vmatrix}}=\frac{\frac{E_2}{j\omega C}-E_1R-\frac{E_1}{j\omega C}}{-j\omega LR-\frac{j\omega L}{j\omega C}-\frac{R}{j\omega C}}=\frac{10000i}{-10000}=1\angle -90^\circ$$

以下同理

$\boxed{16.9}$



$E_1=5$, $E_2=10$, $E_3=3i$

$I_a(4+1-5i)+I_b=5-5i$

$I_b(2+1+3i)+I_a=10-5i$

$$\Delta=\begin{vmatrix} 5-5i & 1 \\ 1 & 3+3i \end{vmatrix}=(5-5i)(3+3i)-1=29$$

a的係數 b的係數 E的 b的

$$\Delta a=\begin{vmatrix} 5-5i & 1 \\ 10-5i & 3+3i \end{vmatrix}=(5-5i)(3+3i)-(10-5i)=20+5i$$

a的 E的

$$\Delta b=\begin{vmatrix} 5-5i & 5-5i \\ 1 & 10-5i \end{vmatrix}=(5-5i)(10-5i)-(5-5i)=20-70i$$

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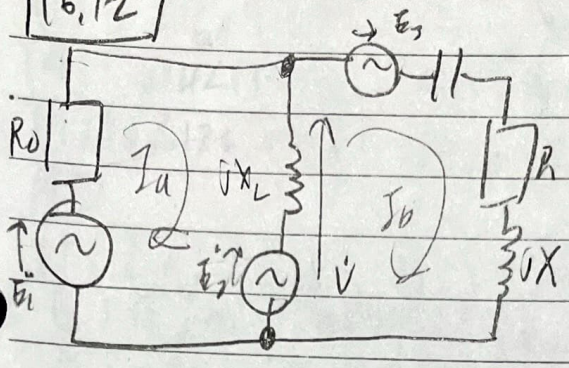
$I_a = \frac{\Delta a}{\Delta} = \frac{20+5i}{29} = 0.71\angle 14^\circ = I_1$

$I_b = \frac{\Delta b}{\Delta} = \frac{20-70i}{29} = 2.5\angle 106^\circ = I_2$

$I_3 = I_a + I_b = 2.6\angle 122^\circ$

$\boxed{6.12}$



$I_a(10+10i) - (10i)I_b = 100 - 100i$

$I_b(5+5i) - (10i)I_a = \cancel{100i - 100i}\ 0$

$\Delta = \begin{vmatrix} 10+10i & -10i \\ -10i & 5+5i \end{vmatrix} = (10+10i)(5+5i) + 100 = 100 + 100i$

$50 + 50i + 50i - 50$

$\Delta_a = \begin{vmatrix} 100-100i & -10i \\ 0 & 5+5i \end{vmatrix} = (100-100i)(5+5i) = 1000$

$\Delta_b = \begin{vmatrix} 10+10i & 100-100i \\ -10i & 0 \end{vmatrix} = (100-100i)(-10i) = -1000 - 1000i$

$I_a = \frac{\Delta a}{\Delta} = 7.07\angle -45^\circ$, $I_b = \frac{\Delta b}{\Delta} = 10\angle 0^\circ$

$\dot{V} = -10i(7.07\angle -45^\circ) = 70.7\angle 45^\circ$